

Quantum field theory as a theorem machine

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- ▶ I've gotten old enough that I regularly make cultural references to before any of you were born.
- ▶ I'm a mathematician, not a physicist.



Physicists have a machine—path integrals, dualities, string theory—that outputs precise predictions about mathematical objects. Predictions that mathematics cannot currently derive on its own.

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My research: take such a prediction and build the mathematics needed to prove it. This is never a formality—it usually requires new structures, and those structures often say something back to the physics.

(I won't compute a cross-section for you. But if you want to know what your QFT is doing to *mathematics*, that's my department.)

Start with a 3d gauge theory: a compact group G_c acting on a complex representation N , and the induced σ -model into $T^*N \cong \mathbb{H} \otimes_{\mathbb{C}} N$. Call it $\mathcal{T}(G, N)$; it has $\mathcal{N} = 4$ supersymmetry.

With this much SUSY, there are two topological twists Q_A and Q_B .

In a twisted 3d theory, the local operators form a **commutative algebra** with a **Poisson bracket** (the secondary product): the functions on the moduli space of vacua.

Two twists $\Rightarrow$ two Poisson varieties:

$$\mathfrak{M}_H = \text{Spec } Z_B(S^2) \quad \mathfrak{M}_C = \text{Spec } Z_A(S^2)$$

the **Higgs** and **Coulomb** branches.

The two branches are not created equal:

- ▶ $\mathfrak{M}_H$ is classical: the hyperkähler quotient $\mathbb{H}^n // G_c$ (Hitchin–Karlhede–Lindström–Roček, 1987). Mathematicians digested this decades ago.
- ▶ $\mathfrak{M}_C$ receives quantum corrections from monopoles. For ~ 20 years, physicists computed examples of a space with **no mathematical definition**.

Braverman–Finkelberg–Nakajima (2016): a rigorous definition, via a convolution algebra built from a moduli space of monopoles. New mathematics, forced into existence by physics.

Example: $G = U(1)$ acting on $N = \mathbb{C}^n$ gives $\mathfrak{M}_C \cong \mathbb{C}^2 / \mathbb{Z}_n$.

These spaces include nilcones of semisimple Lie algebras, and behave like a new class of “semisimple Lie algebras,” vastly generalizing the Killing classification.

3d mirror symmetry

(Intriligator–Seiberg, 1996): pairs of theories $\mathcal{T}, \mathcal{T}^\dagger$ equivalent in a way that swaps $Q_A \leftrightarrow Q_B$. In particular:

$$\mathfrak{M}_H(\mathcal{T}) \cong \mathfrak{M}_C(\mathcal{T}^\dagger) \quad \mathfrak{M}_C(\mathcal{T}) \cong \mathfrak{M}_H(\mathcal{T}^\dagger)$$

- ▶ To a physicist: a duality. To a mathematician: an outrageous list of conjectures pairing spaces with no visible relationship.
- ▶ From this perspective, 3d mirror symmetry generalizes the Langlands duality of semisimple Lie algebras.

Theorem (Braden–Licata–Proudfoot–W.)

For many such dual pairs, the categories $\mathcal{O}$ —categories of representations tailored to the geometry of the two branches—are Koszul dual.

Representation theory on one side determines representation theory on the other. The physicists consider this obvious; the proof is ~ 200

Where this thread is now:

- ▶ **Functoriality:** gauging on the physics side becomes a Hamiltonian reduction operation between Coulomb branches; this proves, e.g., that $\overline{T^*(G/U)}$ for $SL(n)$ is a Coulomb branch (conjecture of Bourget et al.).
- ▶ **Coherent sheaves:** explicit quiver-style descriptions of the categories of sheaves on resolved Coulomb branches—in physics language, categories of boundary conditions—with tilting bundles generalizing Bezrukavnikov's noncommutative Springer resolution.

These categories are exactly what is needed to make Aganagic's string-theoretic knot homology into algebra.

... which brings me to knots.

Witten (1989): expectation values of Wilson loops in Chern–Simons theory are knot invariants; $SU(2)$ with the fundamental representation gives the Jones polynomial.

Coloring a knot K by $\bigwedge^m \mathbb{C}^N$ for all m, N packages into the two-variable **HOMFLYPT polynomial**: a function of q and $g = q^N$.

Garoufalidis–Lauda–Lê: for fixed K , these invariants satisfy recursion relations in the coloring m —the “deformed $\hat{A}$ -polynomial.”

Physics question (’t Hooft): what happens in the planar limit?

$$N \rightarrow \infty, \quad q \rightarrow 1, \quad g = q^N \text{ fixed}$$

Gopakumar–Vafa: large- N Chern–Simons on S^3 is a topological string on the resolved conifold. A knot adds a brane on its conormal Lagrangian L_K .

- ▶ The brane has its own invariant: **knot contact homology** and its augmentation variety (Ekholm, Ng, ...)—defined by counting holomorphic curves, with no reference to Chern–Simons whatsoever.

Prediction (Aganagic–Vafa)

The augmentation variety of K is the classical limit of the recursion relations satisfied by its HOMFLYPT polynomials.

Two objects, two communities, no known map between them. This is exactly the pattern from the start of the talk.

Idea (after Gaiotto–Kannagi–Sanjurjo): treat the coloring m as a **formal variable**. Then $\mu = q^m$ and the shift $\lambda: m \mapsto m - 1$ generate a quantum torus, $\lambda\mu = q\mu\lambda$, acting on a module built from skein theory: the **HOMFLYPT difference module**.

One honest calculation. The MOY relations give, for the unknot:

$$\bigcirc^m = \frac{q^{N-m+1} - q^{-N+m-1}}{q^m - q^{-m}} \bigcirc^{m-1}$$

Substituting $g = q^N$, $\mu = q^m$ (no N , no m !):

$$\bigcirc^m = \frac{qg\mu^{-1} - q^{-1}g^{-1}\mu}{\mu - \mu^{-1}} \lambda \bigcirc^m$$

So $\mu - \mu^{-1} - \lambda(qg\mu^{-1} - q^{-1}g^{-1}\mu)$ annihilates the unknot: its deformed $\hat{A}$ -polynomial, with no computation of any knot invariant.

Theorem (W.–Zaimi, 2026)

At $q = 1$, the HOMFLYPT difference module **is** the degree-0 abelianized knot contact homology of K .

- ▶ This does not fully prove Aganagić–Vafa, but reduces it to sharply stated algebra conjectures (injectivity of an evaluation map). Here is exactly what remains.
- ▶ The proof is purely algebraic: diagram calculus. No holomorphic curves, no path integrals. Elementary tools, non-elementary conclusion.

This paper owes a large debt to conversations with Davide Gaiotto—the kind of accident that only happens in this building.

The same shape, twice:

- ▶ Physics predicted equalities between independently defined mathematical objects: Higgs $\leftrightarrow$ Coulomb; recursions $\leftrightarrow$ augmentation varieties.
- ▶ Proving them required inventing structures—BFN construction, symplectic duality, difference modules—that now have lives of their own.

And the machine keeps producing: Aganagić predicts knot homologies for $\mathfrak{gl}(m|n)$ — $\mathcal{N} = 2$ physics, matrix factorizations—with no mathematical construction yet.

Physics is decades ahead of the proofs. Catching up is a full-time job, with room for help.